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Date: 09/04/2026

Duration: 3 hours

FINAL ASSESSMENT

II YEAR – SEMESTER 4

CSE23AE204 - PCP III

(B.Tech. E01,E02,E03,E05,E06)

SET C

Question 1: Problem Statement

You are developing a **Smart Home Device Control System** for a modern IoT-based home.

The system processes **one device command at a time**, provided as a **space-separated string**.

Each device belongs to a specific category and performs operations differently.

The system must:

- Identify the device type from input
- Perform the device operation based on its configuration
- Use **abstraction and polymorphism** to support multiple device types
- Assign a unique device ID using a **static variable**
- Return the final computed output of the device operation

Device Operation Rules

Smart Light

- Power value represents brightness per level
- Final output is calculated based on brightness and level

Smart Fan

- Power value represents speed power
- Final output is calculated based on power and speed level

Smart AC

- Power value represents cooling capacity
- Final output is calculated based on capacity and mode level

Example 1

Input

```
{ "device": "FAN Fan1 60 3" }
```

Output

180

Explanation

The fan computes the operation result based on its power value and speed level.

Question 2: Problem Statement

You are given a JSON object containing a list of items and the capacity of a knapsack.

```
{ "capacity": W, "items": [[value, weight], ...] }
```

Each item has:

- value → value of the item
- weight → weight of the item

Unlike the 0/1 knapsack problem, **you are allowed to take fractions of an item**.

Your task is to maximize the **total value that can be placed in the knapsack**.

Strategy:

Select items based on the **highest value-to-weight ratio first**.

Input

```
{
  "capacity": 50,
  "items": [[60,10],[100,20],[120,30]]
}
```

Output

240.0

Explanation:

Take items greedily:

- Take item1 → value = 60
- Take item2 → value = 160
- Take **20 weight of item3** → value = 80

Total value = **240**

Question 3: Problem Statement

You are given **two integer arrays**:

- $\text{gas}[i] \rightarrow$ amount of gas available at station i
- $\text{cost}[i] \rightarrow$ gas required to travel from station i to station $i+1$

All gas stations form a **circular route**.

If you start at station i , you:

1. Fill $\text{gas}[i]$
2. Travel to the next station using $\text{cost}[i]$ fuel.

Your task is to determine the **starting gas station index** from which you can travel the **entire circular route exactly once**.

If completing the circuit is **not possible**, return **-1**.

Example 1

Input: $\text{gas} = [1,2,3,4,5]$, $\text{cost} = [3,4,5,1,2]$

Output: 3

Explanation

Start at station **3**: tank = 4

$4 - 1 = 3 \rightarrow +5 = 8$

$8 - 2 = 6 \rightarrow +1 = 7$

$7 - 3 = 4 \rightarrow +2 = 6$

$6 - 4 = 2 \rightarrow +3 = 5$

$5 - 5 = 0 \rightarrow +4 = 4$

The circuit completes successfully.

Question 4: Problem Statement

You are given an array $\text{energy}[]$ representing energy gain or loss at each step.

- Positive values $\rightarrow$ energy gain
- Negative values $\rightarrow$ energy loss

You must choose a **continuous segment (subarray)** such that the **total energy is maximized**.

Return the **maximum possible sum** of such a segment.

Example 1

Input: $\text{energy} = [-2, 1, -3, 4, -1, 2, 1, -5, 4]$

Output: 6

Explanation:

Subarray $[4, -1, 2, 1] \rightarrow \text{sum} = 6$

Question 5: Problem Statement

You are given a list of ropes where each rope has a certain **length**.

You want to connect all ropes into **one single rope**.

When you connect two ropes of lengths a and b :

$\text{cost} = a + b$

After connecting them:

new rope length = $a + b$

This newly formed rope can again be used to connect with other ropes.

Your goal is to determine the **minimum total cost required to connect all ropes into one rope**.

Key idea:

To minimize cost, you should **always connect the two smallest ropes first**.

Example 1

Input: $\text{ropes} = [4,3,2,6]$

Output: 29

Explanation: Step-by-step:

$2 + 3 = 5$ cost = 5

ropes $\rightarrow [4,5,6]$

$4 + 5 = 9$ cost = 9

ropes $\rightarrow [6,9]$

$6 + 9 = 15$ cost = 15

Total cost:

$5 + 9 + 15 = 29$